

Interpretable Safety Constraint Inference in Autonomous Driving using Modal Logical Neural Networks

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Abstract—Autonomous decision-making systems operating in safety-critical environments must reconcile local behavioral responses with long-term planning objectives under dynamically changing conditions. This work investigates the Modal Logical Neural Network (MLNN) as an interpretable neuro-symbolic framework for learning safety-relevant modal constraints in autonomous-driving scenarios. A counterfactual kinematic simulation environment was developed to generate interactions between an Ego vehicle and pedestrian agents under multiple behavioral policies and intervention branches. Safety-relevant predicates describing interaction geometry, control states, and contextual agent behavior were evaluated under modal logical templates representing collision and stopping constraints. Experimental results showed that the learned clause structures emphasized interpretable safety-relevant predicates that are aligned with established safety requirements, including stopping behavior, crosswalk occupancy, collision conditions, and pedestrian intent. Comparison against conventional MLP baselines further demonstrated that the MLNN achieved competitive performance while preserving an explicit modal-logical representation of the learned constraints. These results highlight the potential of modal neuro-symbolic reasoning for interpretable safety-constraint inference in autonomous systems.

Index Terms—autonomous driving, neuro-symbolic learning, modal logical neural networks, interpretable machine learning, safety constraint inference

I. INTRODUCTION

Recent advances in artificial intelligence have enabled autonomous systems to operate in increasingly complex and dynamic environments, where autonomous decision-making systems must continuously reconcile local behavioral demands with long-term planning objectives while reasoning about safety constraints, contextual interactions, and uncertain future states. Existing approaches generally fall into two broad categories. Data-driven methods, such as deep neural networks and reinforcement learning systems, are effective at extracting complex patterns from large-scale datasets but often exhibit limited interpretability and transparency in safety-critical settings [1]. In contrast, symbolic and rule-based approaches provide transparent and structured reasoning mechanisms but often struggle to scale to highly dynamic and high-dimensional

environments [2]. These complementary limitations have motivated the development of hybrid neuro-symbolic approaches that seek to combine the adaptability of statistical learning with the interpretability of symbolic reasoning [2].

Algorithmic autonomous decision-making frameworks have explored a variety of approaches for integrating behavioral reasoning with motion planning. EPSILON proposes a hierarchical planning framework that combines semantic scene understanding with trajectory generation in order to improve decision consistency in dynamic driving environments [3]. MARC further extends this direction through a modular tree-based reasoning architecture that coordinates local behavioral reasoning with long-term planning objectives [4]. These approaches provide structured and modular planning capabilities, but they often face challenges related to scalability, computational complexity, and the integration of highly dynamic contextual information within a unified interpretable framework.

Neural-network-based approaches increasingly rely on learned predictive representations and auxiliary safety mechanisms to improve navigation reliability in dynamic environments. SafetyNet introduces safety-oriented fallback procedures that supervise learned navigation policies to improve robustness during autonomous operation [5], while occupancy-prediction approaches learn representations of surrounding agents and environmental states to support motion planning under uncertainty [6]. More recently, DTPP proposed a differentiable tree-policy planning framework that jointly trains motion prediction and planning-cost evaluation modules within a unified architecture. Unlike earlier hierarchical approaches, DTPP incorporates a query-centric Transformer-based prediction model inspired by Vaswani et al. [7], enabling ego-conditioned motion prediction and contextual reasoning over interacting agents [8]. Reinforcement learning (RL) approaches similarly combine learned policies with external planning or guidance modules; for example, Goldsztejn and Brafman bridge symbolic planning with learned behaviors [9], whereas Brito et al. integrate action-aware guidance mechanisms to improve navigation reliability [10]. While these methods demonstrate strong performance in complex environments,

they often rely on multi-stage architectures, fallback safety layers, or distributed reasoning mechanisms that can reduce interpretability and complicate the integration of explicit safety constraints within a unified reasoning framework.

Neuro-symbolic logical neural networks combine the representational flexibility of neural learning with the interpretability and formal structure of symbolic logic. Logical Neural Networks (LNNs) incorporate logical operators directly into neural architectures, enabling explicit predicate-level reasoning while preserving compatibility with gradient-based learning [11]. Building upon this framework, the Modal Logical Neural Network (MLNN) extends logical reasoning into modal and counterfactual domains through possible-world semantics and accessibility relations [12], making it particularly attractive for autonomous decision-making tasks involving safety constraints, uncertainty, and temporally evolving interactions.

The proposed approach investigates the MLNN as a unified neuro-symbolic framework for autonomous decision-making tasks. Unlike existing multi-stage or modular architectures that distribute reasoning across separate planning, prediction, or safety mechanisms, the MLNN directly models safety-relevant constraints within an interpretable predicate-based reasoning structure. Its architecture preserves explicit logical relationships while supporting gradient-based learning and modal reasoning over contextual and counterfactual environments. Within the proposed framework, the MLNN is trained on a contextualized autonomous-driving simulation environment designed to capture temporally evolving interactions, behavioral variation, and safety-critical conditions. The main contributions of this work are summarized as follows:

- Investigation of the MLNN within the context of autonomous-driving safety and decision-making tasks, which, to the best of the authors' knowledge, has not been previously explored in the literature
- Development of a counterfactual autonomous-driving simulation framework for evaluating modal neuro-symbolic reasoning under safety-critical conditions
- Demonstration of an interpretable predicate-based approach for learning safety-relevant modal constraints within dynamic autonomous-driving environments

The remainder of this paper is organized as follows. Section II presents the MLNN theoretical background, Section III describes the proposed methodology and autonomous-driving simulation framework, Section IV presents the experimental evaluation and analysis, and Section V concludes the paper with limitations and future research directions.

II. PRELIMINARIES

Logical Neural Networks (LNNs) [11] and Modal Logical Neural Networks (MLNNs) [12] combine neural learning with explicit logical reasoning through syntax-tree-like architectures that preserve a direct correspondence between logical expressions and learnable pathways. Truth is represented as an interval bounded by learned lower and upper values, $[L, U]$, given by

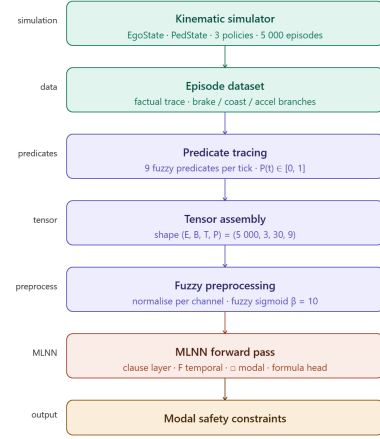


Fig. 1. Overview of the proposed framework

$$L_{out} = f\left(\sum_i L_i w_i - \beta_i\right) \quad (1)$$

$$U_{out} = f\left(\sum_i U_i w_i - \beta_i\right), \quad (2)$$

where f is typically a linear threshold function that guarantees a set of possible results in the range $f \in [0, 1]$.

The MLNN extends the LNN framework through the introduction of modal operators and possible-world semantics. Rather than reasoning over a single observed state, modal operators evaluate logical constraints across sets of reachable states, enabling reasoning about necessity $\square$ and possibility $\diamond$ within contextual and counterfactual environments. This capability makes the MLNN particularly suitable for applications involving uncertainty, temporal evolution, and safety-critical decision making.

In autonomous-driving applications, modal reasoning provides a natural framework for expressing safety requirements across multiple operating conditions and future outcomes. Consequently, safety constraints can be evaluated not only with respect to a single observed state but also across a range of reachable contextual scenarios.

III. METHODOLOGY

The proposed experimental framework, used to evaluate the MLNN in a safety-relevant autonomous-driving environment, is described in this section. A counterfactual kinematic simulation was developed to generate temporally evolving interactions between Ego and pedestrian agents under varying behavioral conditions. These interactions were represented through fuzzy logical predicates and evaluated under modal logical templates. Fig. 1 illustrates the overall experimental workflow, from simulation generation and predicate extraction to MLNN processing and modal-template evaluation.

TABLE I
SIMULATION PARAMETER VARIATION CATEGORIES

Parameter	Distribution
Crosswalk x-center	$U(12, 22)$ m
Crosswalk width	$U(3, 8)$ m
Crosswalk y-extent	Fixed ± 10 m
Ego initial speed	$U(4, 13.9)$ m/s
Ego spawn x	Fixed 0
Pedestrian spawn y	$U(-14, -8)$ m
Pedestrian target y	$U(+8, +14)$ m
Pedestrian lateral x	$x_{ew} + U(-2, +2)$
Pedestrian speed	$U(0.8, 1.8)$ m/s
Pedestrian acceleration phase	$U(0.5, 2.0)$ m
Ego policy	Cautious 60%, Late 20%, Aggressive 20%
Entry side	Left/Right (50/50)
Weather preset	10 categories

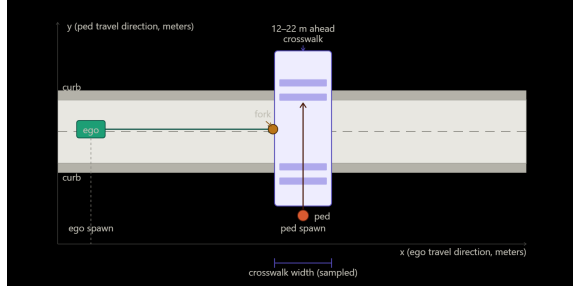


Fig. 2. Autonomous-driving simulation environment

A. Kinematic Simulation

A Python-based kinematic simulation was developed to generate a controlled 2D autonomous-driving scenario involving an Ego vehicle and a pedestrian. Each episode was sampled independently from a fixed set of distributions and produced one factual trace and three counterfactual branches corresponding to alternative Ego intervention policies. Table I summarizes the simulation parameter space.

For the purpose of counterfactual world segmentation, each episode generated four traces: one factual trace of 60 ticks and three branch traces of 30 ticks each corresponding to interventions applied to the Ego’s braking, coasting, and acceleration coefficients at the recorded fork point. The fork point was defined as the first tick during the factual trace for which *PedestrianIntentToCross* evaluated as true.

An episode began when the simulator detected a safety-relevant crosswalk encounter during the pre-simulation wait phase, defined by

$$PedestrianIntentToCross \wedge dist(ego, ped) < 20 \text{ m} \quad (3)$$

This condition indicated that the pedestrian was approaching the crosswalk while the Ego vehicle was sufficiently close for a safety-relevant interaction. An episode terminated when either the pedestrian reached the far curb (*PedestrianResolved*) or the Ego vehicle moved more than 40 m beyond the pedestrian. The factual trace was capped at a maximum duration of 60 ticks regardless of resolution state. Figure 2 illustrates the simulation environment and counterfactual branching structure.

The Ego vehicle spawned at the origin, as a point mass, and traveled along the positive x-axis. Its initial speed was

sampled from $U(4.0, 13.9)$ m/s, while throttle and brake coefficients were initialized to zero. The Ego then coasted at its sampled speed until the selected braking policy engaged (see Ego policy in Table I). Pedestrian spawning conditions were parameterized relative to the sampled crosswalk geometry. The pedestrian’s lateral position along x was defined as the crosswalk center plus a small uniform offset, while its initial y-position and velocity were sampled according to the selected entry side and motion policy. Crosswalk geometry was regenerated independently for each episode. The crosswalk center was sampled from $U(12, 22)$ m ahead of the Ego spawn location in order to guarantee that aggressive-policy Egos could reach the crosswalk before stopping. Crosswalk width was sampled across urban (3-4m), standard (4-6m), and wide (6-8m) categories, while the crosswalk y-extent was fixed to cover the full road width and footpath margins. Angled crosswalks were not modeled.

At the recorded fork tick, both the Ego and pedestrian were reset to their original spawn states before executing the corresponding counterfactual branch policy. The Ego speed was restored to its sampled initial value, while the pedestrian’s position and velocity were reset to their original sampled conditions. Each branch then executed forward for 30 ticks under its assigned intervention policy. This reset procedure prevented scenarios in which the pedestrian had already crossed a substantial portion of the road before the Ego reached the crosswalk, a condition that previously produced artificially suppressed collision rates in the acceleration branch.

B. Predicates

Nine fuzzy logical predicates were selected to evaluate the MLNN under two safety-relevant modal logical templates, F2 and F3. A preliminary template, F1, was discarded because it remained trivially satisfied under the simulation conditions. For readability, *PedInCrosswalk* and *Stopped* denote the *PedestrianInCrosswalk* and *EgoStopped* predicates, respectively, while *LowTTC* denotes a *TimeToCollision* value below a predefined safety threshold.

$$F2 : \Box(Collision \rightarrow \neg \Diamond Accel) \quad (4)$$

$$F3 : \Box(PedInCrosswalk \wedge LowTTC \rightarrow \Diamond Stopped) \quad (5)$$

Template F2 expresses the constraint that collision states should not permit future acceleration behaviors across reachable branches, whereas F3 encodes the expectation that the Ego vehicle should eventually stop when a pedestrian occupies the crosswalk under low time-to-collision conditions. Here, $\Box$ and $\Diamond$ denote modal necessity and eventuality operators evaluated over counterfactual branch traces. These templates were designed to represent simplified yet representative safety constraints commonly encountered in pedestrian-aware autonomous-driving scenarios. Specifically, F2 captures post-collision behavioral consistency, whereas F3 captures preventive safety behavior in pedestrian-crossing situations.

The predicates were grouped into four broad categories: continuously tracked interaction geometry (*DistanceToPedestrian*, *TimeToCollision*), discrete interaction events (*PedestrianInCrosswalk*, *EgoInCrosswalk*, *Collision*), Ego control states (*EgoBraking*, *EgoAccelerating*, *EgoStopped*), and pedestrian behavioral context (*PedestrianIntentToCross*).

DistanceToPedestrian ranged from near zero at collision to approximately 40 m at episode initialization. After normalization and fuzzy sigmoid mapping, lower distances produced higher fuzzy truth values corresponding to pedestrian proximity. Similarly, *TimeToCollision* was represented as a continuous variable with a sentinel value of 999s when the Ego and pedestrian trajectories diverged. After normalization, low TTC values mapped to higher risk-associated fuzzy activations, whereas large TTC values produced lower activations.

The remaining predicates were Boolean. *PedestrianInCrosswalk* and *EgoInCrosswalk* tracked whether the corresponding bounding boxes overlapped with the crosswalk region, while *Collision* activated on the first tick during which the Ego and pedestrian bounding boxes intersected. The use of the possibility operator in F2 was important because it allowed collisions to occur within reachable branch traces without requiring collisions across all branches simultaneously. A necessity-based collision constraint would have been largely unattainable under the simulation conditions and would therefore have rendered F2 trivially satisfied.

EgoBraking and *EgoAccelerating* were used to characterize the Ego vehicle’s control state throughout each rollout. Because braking and acceleration were mutually exclusive behaviors under the simulation logic, they also served as a coarse validation check during initial simulator development. *EgoBraking* activated when brake input exceeded a fixed threshold, whereas *EgoAccelerating* activated when throttle input exceeded a threshold while brake input remained zero. Since *EgoAccelerating* appears in negated form within F2, the template evaluates whether collision conditions imply the absence of continued acceleration. The final control-state predicate, *EgoStopped*, activated when Ego speed fell below a small threshold and was used in F3 to evaluate whether the Ego vehicle eventually stopped during the rollout.

As discussed previously, *PedestrianIntentToCross* served as both a spawning and branching condition. It was initially incorporated into the preliminary template

$$F1 : Intent \rightarrow \Diamond Brake$$

but was ultimately discarded because episode recording began only after *PedestrianIntentToCross* evaluated as true. As a result, the predicate exhibited near-zero variance throughout the dataset, causing F1 to collapse into a trivial brake-activation condition rather than a meaningful modal relationship. Although the template was abandoned, the predicate itself remained useful as contextual information associated with safety-relevant interaction scenarios.

Collectively, the selected predicates capture interaction geometry, agent context, and vehicle control states that are

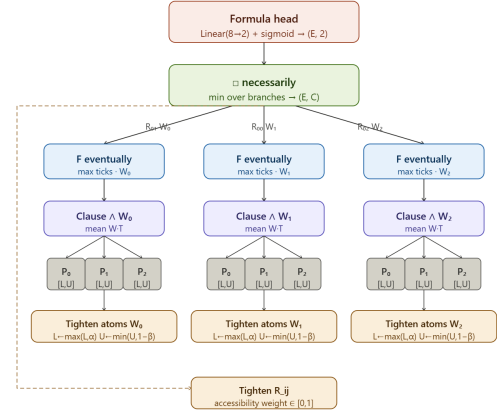


Fig. 3. MLNN processing flow from predicate inputs to modal-template evaluation

commonly used in autonomous-driving systems to characterize collision risk and safety-critical behavior [5], [6], [8], [13]. This combination provides an interpretable representation of pedestrian-vehicle interactions suitable for evaluating modal neuro-symbolic reasoning within the proposed simulation environment.

C. MLNN Processing Pipeline

Figure 3 provides a detailed view of the internal MLNN processing pipeline, complementing the high-level experimental workflow shown in Fig. 1. For each episode, the model reads three branch matrices, each containing 30 time steps and 9 predicate values corresponding to the three counterfactual branch traces. These matrices are stacked along the branch axis to produce a $(3, 30, 9)$ episode-level tensor. Repeating this process over 5000 episodes yields the full input tensor $T \in \mathbb{R}^{5000 \times 3 \times 30 \times 9}$, where the dimensions correspond to episode, branch, time, and predicate, respectively.

The continuous predicate channels were normalized independently across the dataset prior to training. This step was necessary to prevent saturation effects during fuzzy truth-value generation and to preserve useful gradient information. A fuzzy sigmoid transformation,

$$\sigma(10(2x - 1)),$$

was then applied to all predicate values, mapping them to the range $[0, 1]$. Boolean predicates naturally mapped to strong truth values near the interval boundaries, whereas continuous predicates produced smoothly varying fuzzy activations. Tensor T therefore represents the fuzzy truth value associated with each predicate at a particular time step, branch, and episode.

The tensor T is passed to the MLNN clause layer, where each predicate value is interpreted as a fuzzy truth value in the interval $[0, 1]$. The clause layer learns a weighted conjunction of predicates while preserving bounded truth values in the same interval. The output of clause C_i at episode e , branch b , and time step t is given by

$$v(C_i, e, b, t) = \frac{1}{P} \sum_p W_{ip} v(P_p, e, b, t). \quad (6)$$

Evaluating Eq. 6 across all episodes, branches, time steps, and clause cells produces the clause tensor $C[E, B, N_t, I]$, where E , B , N_t , and I denote the episode, branch, temporal, and clause dimensions, respectively.

The clause tensor is subsequently processed by the temporal and modal operators shown in Figure 3. The temporal operator $\diamond$ aggregates clause activations across the time dimension, allowing the network to evaluate eventuality conditions over a rollout. The modal operator $\square$ then aggregates the resulting values across counterfactual branches, enforcing necessity over the reachable branch set. Finally, the formula head maps the resulting clause representations to the target modal-template satisfaction scores used during training.

IV. EVALUATION

Early experiments revealed four primary limitations that prevented effective learning. First, the combination of unnormalized continuous predicates and a high-sharpness fuzzy sigmoid ($\beta = 50$) caused widespread activation saturation, producing nearly constant formula-satisfaction scores. Second, several simulation-design choices, including branch timing and crosswalk placement, substantially reduced the frequency of informative collision scenarios. Third, preliminary modal templates were found to be either vacuous or unlearnable; in particular, F1 exhibited near-zero variance because its antecedent served as an episode-triggering condition. Finally, product t-norm aggregation and MSE-based optimization produced weak gradient signals, motivating the adoption of mean t-norm aggregation, masked binary cross-entropy loss, and a lightweight prediction head.

After these modifications, training stabilized and the resulting model achieved Pearson correlations of $r=0.428$ and $r=0.548$ for F2 and F3, respectively. The final configuration (Run F) was subsequently used for all analyses reported below.

Template F2. The final model configuration was analyzed through the learned formula-head correlations and clause-cell activations. Pearson correlation was used to evaluate the relationship between predicted and target template satisfaction scores, while the learned clause and predicate weights were examined to assess the interpretability of the resulting modal constraints. Representative clause-cell weights are summarized in Table II.

The antecedent of F2 was active in 8.6% of episodes (430/5000), providing a sufficiently informative population for evaluating the collision-related constraint. Because both antecedent and consequent varied across episodes and branches, the template remained non-vacuous and suitable for learning.

This interpretation is supported by the learned formula-head weights. Both the C0 and C4 weights from the F2 formula head were negative: -1.209 and -1.239, respectively. This confirms that high clause activation predicts low F2 satisfaction. From the values given in C4, the Ego and the pedestrian are both in the crosswalk, a collision has occurred, braking is

TABLE II
LEARNED PREDICATE WEIGHTS ACROSS REPRESENTATIVE CLAUSE CELLS
AFTER RUN F

Predicate	C ₀	C ₂	C ₄
EgoInCrosswalk	0.768	0.864	0.744
EgoStopped	0.750	0.796	0.704
PedestrianIntentToCross	0.690	0.780	0.693
DistanceToPedestrian	0.615	0.719	0.681
Collision	0.566	0.699	0.651
EgoBraking	0.553	0.640	0.618
PedestrianInCrosswalk	0.545	0.549	0.586
TimeToCollision	0.538	0.403	0.479
EgoAccelerating	0.359	0.350	0.376

present, and intent is activated. The moderate EgoAccelerating weight likely reflects the consequent’s negation, as would be required by the formula, and, indeed, the negative head weight signals that the formula head learned that when C4 is highly active, F2 satisfaction is lower. This would follow from the observation that C4 activates strongly in the aggressive-policy episodes, where acceleration and collision would occur in violation of F2.

Template F3. The compound antecedent was active in 50% of episodes. This high rate reflects the fact that both conditions were jointly satisfied when a cautious or late-reaction Ego approaches an active crosswalk. The consequent was satisfied in those episodes in which the brake branch produced stopping behavior and it was violated in aggressive Ego episodes in which the ego neither braked nor stopped across any branch. This strongly suggests a non-vacuous template.

F3 earned an r score of 0.548, the higher of the two, and the only one to exceed the 0.5 target. This is likely attributable to the higher share of correlated episodes and the greater predicate variance. Its plateau after 500 epochs indicates that the correlation likely stabilized, ruling out overfitting.

The F3 formula head similarly gave negative weights to C0 and C4: -1.086 and -1.100, respectively. C0 is interpretable as a joint crosswalk occupancy clause: EgoInCrosswalk and EgoStopped are the dominant weights, which indicates that it activates strongly when the Ego is in the crosswalk and has stopped. The relatively high PedestrianIntentToCross weight adds the pedestrian’s approach as context, and the moderate TimeToCollision weight perhaps reflects the LowTTC antecedent condition. Again, the negative head weight indicates that when C0 is highly active the formula head predicts lower F3 satisfaction. This is plausible since the episodes in which the Ego has stopped in the crosswalk with the pedestrian present are the collision-adjacent episodes, i.e., those in which the aggressive Ego world shows the Ego failing to stop. The clause therefore activates on the dangerous but resolved (i.e., no collision) episode profile. This induces the head to observe that high activation of this clause is correlated with the marginal episodes in which F3 varies rather than the unambiguously safe episodes in which the Ego stops reliably across all branches.

TABLE III
MLNN AND MLP BASELINE CORRELATION SCORES FOR TEMPLATES $F2$
AND $F3$

Model	F2	F2	F3	F3
	Run 1	Run 2	Run 1	Run 2
MLNN	0.428	0.428	0.548	0.548
MLP-A ($H = 8$)	0.148	0.174	0.612	0.597
MLP-B ($H = 810$)	0.294	0.592	0.567	0.570
MLP-C ($H = 32$)	0.429	0.266	0.602	0.549

A. Baseline Comparison

To benchmark the MLNN against conventional neural architectures, three MLP baselines with varying input representations and hidden-layer capacities were evaluated. The comparison was designed to assess whether the MLNN’s modal-logical structure provides an advantage when learning safety-relevant constraints under different levels of antecedent participation. MLP-A employed a compact predicate-level architecture ($9 \times 8 \times 2$), while MLP-B ($H = 810$) and MLP-C ($H = 32$) increased the model capacity. Two input representations were considered. The modal representation consisted of nine episode-level features obtained after aggregating the predicate observations through the MLNN modal operators, whereas the flattened representation contained the full episode tensor ($3 \times 30 \times 9 = 810$) reshaped into a single feature vector. Run 1 employed a flat-input MLP-B and modal-input MLP-C, whereas Run 2 employed a modal-input MLP-B and flat-input MLP-C. Table III summarizes the resulting Pearson correlation scores. Swapping the input representations between the two higher-capacity architectures enabled the separate examination of representation and model-capacity effects on constraint inference performance.

The two templates represent different learning regimes, with $F2$ characterized by sparse antecedent participation and $F3$ by a substantially larger informative population. This distinction enables the evaluation of model behavior under varying levels of available training signal.

For template $F2$, performance varied considerably across architectures and input representations. The MLNN achieved a correlation of 0.428, outperforming MLP-A across both runs and remaining competitive with the larger-capacity baselines. Notably, MLP-C achieved a comparable correlation under Run 1, whereas MLP-B achieved the highest correlation under Run 2. These results suggest that both model structure and input representation play important roles when informative training examples are relatively sparse. Despite not achieving the highest correlation, the MLNN maintained competitive performance while preserving an explicit modal-logical interpretation of the learned constraint.

For template $F3$, performance differences were substantially smaller, with all architectures achieving moderate-to-strong correlations. This indicates that the larger antecedent participation rate provides sufficient information for both structured and unconstrained models to learn the underlying relationship. Under these conditions, the MLNN no longer exhibits a clear performance advantage, but achieves results comparable to

the MLP baselines while preserving an interpretable logical structure.

Overall, the comparison suggests that the MLNN remains competitive across both sparse and dense informative populations while preserving an explicit modal-logical representation of the learned constraints. Although larger unconstrained architectures can achieve comparable or higher correlations in certain cases, the MLNN provides an interpretable reasoning structure in which learned behaviors can be traced directly to human-readable predicates and modal constraints, making it well suited for safety-oriented reasoning tasks.

V. CONCLUSION

This work investigated the MLNN as an interpretable neuro-symbolic framework for learning safety-relevant modal constraints in autonomous-driving environments. The learned clause structures emphasized stopping behavior, crosswalk occupancy, collision conditions, and pedestrian intent, while remaining competitive with MLP baselines and preserving an explicit modal-logical representation of the learned constraints. Future work will expand the template dictionary, incorporate additional safety standards, and investigate integration with higher-level autonomous planning systems.

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